Mittuniversitetet
MID SWEDEN UNIVERSITY

Försättsblad tentamen / Examination cover

2707183

Anonymitetskod / Anonymous code		
S - 0003 - SWF		
Kurskod / Course code	Provkod/ Test code	Tentamensdatum / Examination date
MA080G	T102	2026-01-05
Kursnamn / Course name		
Matematik GR (B), Kryptografi		
Provnamn / Test name		
Skriftlig tentamen		

Skriv din anonymitetskod på varje inlämnat papper
Write your anonymous code on each sheet submitted

Casio fx-9750 – miniräknare

Sätt ett kryss (x) för varje inlämnad uppgift
Use an x to indicate which questions has been submitted

Markera nedan med X / Mark below with an X		Poäng / Credit	Lärarens anteckningar / Teacher's notes	Markera nedan med X / Mark below with an X		Poäng / Credit	Lärarens anteckningar / Teacher's notes
1	X	4		16			
2	X			17			
3	X	6,5		18			
4	X	3		19			
5	X	3		20			
6	X	1		21			
7				22			
8				23			
9				24			
10				25			
11				26			
12				27			
13				28			
14				29			
15				30			
Poängsumma / Points		17,5	Betyg / Grade B	Lärarsign./ Teachers sign		M	

Fylls i av tentamensvakt / To be filled in by the invigilator

Antal lösa blad/ No. of sheets submitted	Inlämnad tentamen / Submitted exam	Leg kontroll / Control identification	Sign. tentamensvakt / Sign. invigilator
6	O	OH	BW

2707183

Försättsbladet skall alltid lämnas in även om ingen uppgift behandlats
Examination cover should always be submitted even if no questions are answered

Utskriven 2025-12-29 kl. 09:21:43

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Anonymitetskod/ Anonymus code: S-0003-SWF

Sidnr / Page no: 1

Lärarens anteckning / Teachers note:

1a)

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
0 5 10 15 20 25

K = 3

Answer: K=3, DATA = GDWD ✓ 2

C = GDWD

1b) K E Y
10 4 24

$P_i = C_i - K_i \pmod{26}$

$P_0 = 17 - 10 \equiv 7 \pmod{26} \Rightarrow H$

$P_1 = 8 - 4 \equiv 4 \Rightarrow E$

$P_2 = 9 - 24 \equiv 11 \Rightarrow L$

$P_3 = 21 - 10 \equiv 11 \Rightarrow L$

$P_4 = 18 - 4 \equiv 14 \Rightarrow O$

$P_5 = 20 - 24 \equiv 22 \Rightarrow W$

$P_6 = 24 - 10 \equiv 14 \Rightarrow O$

$P_7 = 21 - 4 \equiv 17 \Rightarrow R$

$P_8 = 9 - 24 \equiv 11 \Rightarrow L$

$P_9 = 13 - 10 \equiv 3 \Rightarrow$

Answer: HELLO WORLD ✓ 2

Skriv ej på baksidan! Do not write on the other side!

Lärarens anteckning / Teachers note:

3a) $\phi(105)$

$105 = 3^1 \cdot 35$

$35 = 5^1 \cdot 7$

$=$

$\phi(3) \cdot \phi(5) \cdot \phi(7)$

$=$

$2 \cdot 4 \cdot 7 = 56$ ×

If n is prime then: $\phi(n) = n - 1$

Answer: 56 ×

0,5

3b) $\gcd(4, 35) = 1$, which means 4 has an inverse, so yes it's a unit in $\mathbb{Z}/35\mathbb{Z}$ ✓

1

3c) $(\mathbb{Z}/12\mathbb{Z}, +) = \{1, 2, \cdots, 12-1\}$

Generators are when $\gcd(a, 12) = 1$, where $a \in \{1, 2, \cdots, 12-1\}$

$\gcd(1, 12) = 1$, $\gcd(2, 12) = 2$, $\gcd(3, 12) = 3$

$\gcd(4, 12) = 4$, $\gcd(5, 12) = 1$, $\gcd(6, 12) = 6$

$\gcd(7, 12) = 1$, $\gcd(8, 12) = 4$, $\gcd(9, 12) = 3$

$\gcd(10, 12) = 2$, $\gcd(11, 12) = 1$ ✓

1

BTW, $\phi(12) = \phi(2^2) \cdot \phi(3^1)$

$= (2^2 - 2^1) \cdot (3^1 - 3^0) = 2 \cdot 2 = 4 = \#$ generators.

Example $g = 5$, $P = 12$, modulus 12.

$5 \cdot 1 \equiv 5$	$5 \cdot 5 \equiv 1$	$5 \cdot 9 \equiv 9$
$5 \cdot 2 \equiv 10$	$5 \cdot 6 \equiv 6$	$5 \cdot 10 \equiv 2$
$5 \cdot 3 \equiv 3$	$5 \cdot 7 \equiv 11$	$5 \cdot 11 \equiv 7$
$5 \cdot 4 \equiv 8$	$5 \cdot 8 \equiv 4$	$5 \cdot 12 \equiv 0$

12 operations.

Answer: generators $= \{1, 5, 7, 11\}$

13th operation goes back to 5.

Lärarens anteckning / Teachers note:

3d)

Smallest k such that $7^k \equiv 1 \pmod{15}$

Mod 15

$7 \equiv 7,\ 7^2 \equiv 4,\ 7^3 \equiv 4 \cdot 7 \equiv 13,\ 7^4 \equiv 4^2 \equiv 1$

Answer: $\text{ord}(7) = 4$ ✓

1

3e) $a = 11,\ n = 40$

If a is a prime and $\gcd(a, n) = 1$

then: $a^{\phi(n)} \equiv 1 \pmod{n}$

$\gcd(11, 40) = 1$

$40 = 2^3 \cdot 5^1$

$\phi(40) = \phi(2^3)\phi(5)$

$= (2^3 - 2^2)(5^1 - 5^0)$

$= 8 \cdot 4 = 32$ ×

So $11^{32} \equiv 1 \pmod{40}$

0,5

3f) $\gcd(17, 43) = 1$, Inverse exists,

$43 = 2 \cdot 17 + 9$
$17 = 1 \cdot 9 + 8$
$9 = 1 \cdot 8 + 1$

EEA
$9 = 43 - 2 \cdot 17$
$8 = 17 - 9$
$1 = 9 - 8$

$1 = 9 - 8 \Leftrightarrow 1 = 9 - (17 - 9) \Leftrightarrow 1 = 9 - 17 + 9$

$\Leftrightarrow 1 = 2(9) - 17 \Leftrightarrow 1 = (2)(43 - 2 \cdot 17) - 17$

$\Leftrightarrow 1 = (2)(43) - (5)(17) \Leftrightarrow$

$1 = 2 \cdot 43 - 5 \cdot 17$

$-5 \equiv 38 \pmod{43}$

Answer: $17^{-1} \equiv 38 \pmod{43}$ ✓

2

3 g)
There is a solution, when $x = 4$,
$14 \cdot 4 = 56 \equiv 12 \pmod{22}$ only this solution? 0,5

4a) $p = 31$, $g = 3$, $a = 7$, $b = 12$

Alice's $K_{pub} = g^a \pmod{p}$

$3^7 \pmod{31}$

$3 \equiv 3 \pmod{31}$

$3^2 \equiv 9$ -11-

$3^4 \equiv 19$ -11-

$3^6 \equiv 19 \cdot 9 \equiv 171 \equiv 16$ -11- ✓

$3^7 \equiv 16 \cdot 3 \equiv 17$ -11- Answer: 17

4b) Same goes to Bob's K_{pub}.
$\pmod{31}$
$3^{12} \equiv (3^6)^2 \equiv 16^2 \equiv 8$ Answer: 8 ✓

4c) $S = h_A$ (or h_B)$^{K_{pr}} \pmod{p}$
So $S = 17^{12} \pmod{31} \equiv 2 \pmod{31}$
or $S = 8^7 \pmod{31} \equiv 2 \pmod{31}$
$\pmod{31}$
$17 \equiv 17$
$17^2 \equiv 10$
$17^4 \equiv 7$
$17^8 \equiv 18$
$17^{12} \equiv 17^8 \cdot 17^4 \equiv 18 \cdot 7 \equiv 126 \equiv 2$ ✓
Answer: $S = 2$

Lärarens anteckning / Teachers note:

Lärarens anteckning / Teachers note:

$p = 103$, $g = 5$, $a = 21$, $C_2 = 34$

5a) $h_A = 5^{21} \pmod{103}$

$(\mod 103)$

$5 \equiv 5$, $5^2 \equiv 25$, $5^4 \equiv 625 \equiv 7$,

$5^8 \equiv 7^2 \equiv 49$, $5^{16} \equiv 49^2 \equiv 2401 \equiv 32$

$5^{21} \equiv 5^{16} \cdot 5^4 \cdot 5 \equiv 32 \cdot 7 \cdot 5 \equiv 1120 \equiv 90$

Answer: $h_A = 90$

5b) $\gcd(90, 103) = 1$, h_A^{-1} exists.

EA) LEEA

$103 = 1 \cdot 90 + 13$ | $13 = 103 - 90$

$90 = 6 \cdot 13 + 12$ | $12 = 90 - 6 \cdot 13$

$13 = 1 \cdot 12 + 1$ | $1 = 13 - 12$

$12 = 12 \cdot 1 + 0$

$1 = 13 - 12 \Leftrightarrow 1 = 13 - (90 - 6 \cdot 13) \Leftrightarrow$

$1 = 13 - 90 + 6 \cdot 13 \Leftrightarrow 1 = (7)(13) - 90 \Leftrightarrow$

$1 = (7)(103 - 90) - 90 \Leftrightarrow 1 = (7)(103) - (8)(90)$

$\Leftrightarrow 1 = 7 \cdot 103 - 8 \cdot 90$

Answer: $h^{-1} = -8 \equiv 95 \pmod{103}$

5c) $m = 34 \cdot (-8) \equiv 37 \pmod{103}$

or

$m = 34 \cdot 95 \equiv 3230 \equiv 37 \pmod{103}$

Lärarens anteckning / Teachers note:

6a) $n = 221$, $e = 5$

$\phi(221) = \phi(13) \cdot \phi(17)$

$= 12 \cdot 16 = \boxed{192}$

221 / 13 ; 17

This because If n is prime, then

$\phi(n) = n - 1$

or

$\phi(221) = 221\left(1 - \frac{1}{13}\right)\left(1 - \frac{1}{17}\right)$

$= 221\left(\frac{12}{13} \cdot \frac{16}{17}\right) = 221\left(\frac{192}{221}\right)$ ✓

$\frac{221}{1} \cdot \frac{192}{221} = \frac{42432}{221} = \boxed{192}$

1

Answer: 192

6b) $5 \cdot d \equiv 1 \pmod{192}$ ✗

6c) $c = m^e \pmod{n}$

$c = 13^5 \pmod{221} \equiv 13$ //

✗